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For Teachers & Departments

# Advanced Statistics for Edexcel A-Level

A Comprehensive Guide to Core, Further and STEP Statistics

Combined Edition • Pearson Edexcel 9ST0 • ISBN 978-1-0666184-0-8

by Isaac Teng — MSci Mathematics & Physics, UCL; Founder, IREducation Inc.

## What this evaluation copy contains

These are complete, unaltered excerpts from the full textbook. Page and chapter numbers are those of the published book, so what you see here is exactly what students receive.

- **Front matter** — Introduction, About the Author, and the full Contents.
- **Chapter 12** — Bayes' Theorem (pp. 215–230).
- **Chapter 17** — Hypothesis Testing for 1 and 2 Samples (pp. 293–310).
- **Appendix B excerpt** — calculator guide Sections B.1–B.3: random numbers, one-variable statistics, correlation and linear regression.
- **Practice Paper 02** — Statistical Inference, with its matching worked answers.

## The full book at a glance

- 466 pages and 21 chapters spanning Core, Further and STEP statistics.
- Three full practice papers with complete worked answers, plus per-chapter answers.
- Complete appendices on calculator use, the Statistical Enquiry Cycle and statistical reference.



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Combined Edition

*A Comprehensive Guide to Core, Further and STEP Statistics*

Isaac Teng Chit Hang

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*A Comprehensive Guide to Core, Further and STEP Statistics*

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**IRMaths by IREducation Inc.**
Website: `irmaths.ireducation.org`

*Second Edition, 2026*
ISBN-10: 1-0666184-0-2
ISBN-13: 978-1-0666184-0-8

While every effort has been made to ensure the accuracy of the mathematics, code, and explanations contained within this book, the publisher and author assume no responsibility for errors or omissions, or for damages resulting from the use of the information contained herein.

# Introduction

This textbook provides a comprehensive foundation for the standalone **Edexcel A-Level Statistics** qualification (specification code **9ST0**). To offer a broad and mathematically rigorous approach to the subject, the text integrates material from across multiple syllabi. It encompasses the core statistics components of standard A-Level Mathematics, concepts introduced in A-Level Further Mathematics (Further Statistics), and the calculus-based probability required for the Sixth Term Examination Paper (STEP).

This combined edition contains the complete sequence of Chapters 1–21, the three practice papers, and their answers.

This second edition includes improved explanations across a range of topics, with additional clarification where students commonly require further support. Some exercises have also been revised or replaced so that they more closely reflect the style, scope, and level of challenge of questions that students may encounter in the examination.

The objective of this book is to guide students systematically from foundational statistical concepts to advanced mathematical problem-solving. By combining core theory, detailed worked examples, and structured exercises, the text aims to develop both conceptual understanding and technical proficiency. Topics range from experimental design and standard hypothesis testing to Analysis of Variance (ANOVA), non-parametric methods, and continuous probability distributions. Throughout the chapters, there is a consistent emphasis on mathematical rigour, precise notation, and logical communication.

Designed to synthesise these overlapping specifications into a single, coherent framework, the book functions equally well as a primary classroom text and as an independent reference manual. Students are encouraged to use this material in conjunction with the IRMaths platform (`irmaths.ireducation.org`), which provides interactive resources, dynamic explanations, and further opportunities for practice and consolidation.

While every effort has been made to ensure the mathematical and typographical accuracy of this text, occasional errors may inevitably occur in a work of this scope. Should you encounter any errors, or if you have suggestions for corrections and improvements, please feel free to contact me at `isaacteng@ireducation.org`.

Thank you for choosing to use this textbook, and I wish you every success in your studies.

***A Note on the Beyond the Syllabus Content:***

*Throughout the text, you will encounter purple* ***Beyond the Syllabus*** *boxes. These sections explore the broader mathematical framework behind standard formulas, offering formal proofs, historical context, and connections to modern disciplines such as machine learning, data science, and Bayesian statistics. While this material falls outside the scope of standard A-Level examinations, it is intended to cultivate deeper intuition, support the critical thinking required for STEP, and serve as preparation for university-level study in quantitative fields.*

# About the Author

Isaac Teng is the founder of IREducation Inc., an ed-tech organization based in Hong Kong SAR. He is also an MSci Mathematics and Physics student at University College London (UCL), and the lead developer for the IREducation platform, including IRMaths.

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# Chapter 12

# Bayes' Theorem

**After completing this chapter you should be able to:**

- Calculate and use conditional probabilities.
- Construct and interpret probability tree diagrams for up to three events.
- Understand and apply Bayes' theorem to calculate posterior probabilities.
- Apply Bayes' theorem to scenarios involving up to three mutually exclusive events.
- Solve complex algebraic problems involving unknown probabilities using Bayes' theorem.

## 12.1 Conditional Probability and Tree Diagrams

In earlier studies of probability, you learned how to calculate the probability of an event occurring given that another event has already occurred. This is known as **conditional probability**.

### 12.1.1 Conditional Probability Fundamentals

The probability of event $A$ occurring, given that event $B$ has already occurred, is written as $P(A|B)$.

**Key Point**

The formula for conditional probability is:

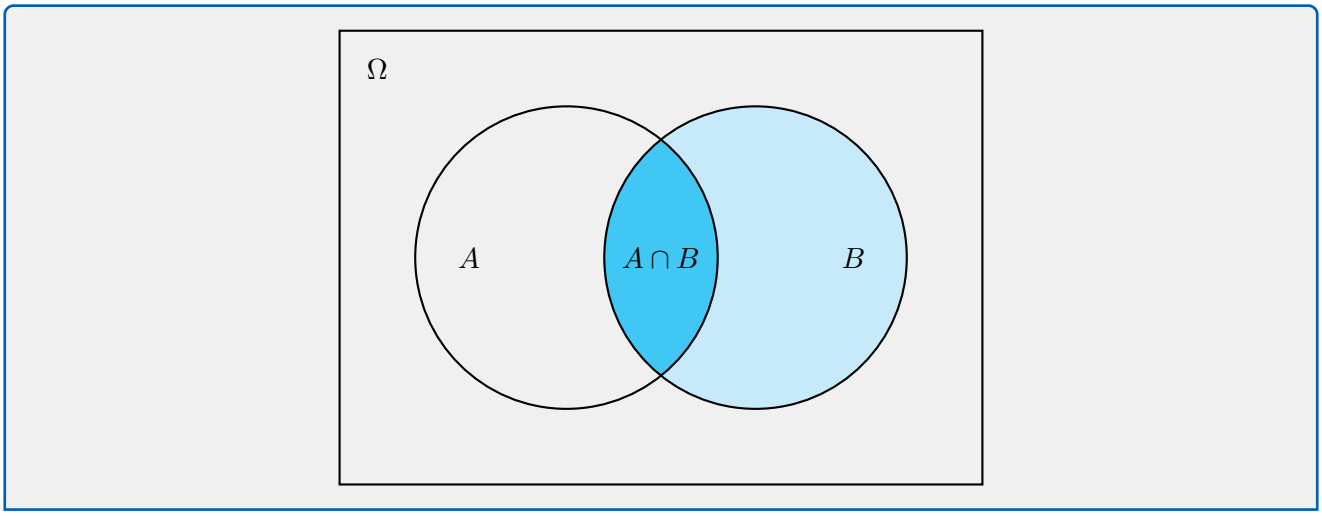
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Rearranging this gives one form of the **multiplication rule** for finding the intersection (the probability of both events happening):

$$P(A \cap B) = P(B) \times P(A|B)$$

Similarly, from the definition of $P(B|A)$, we get another form that is often used:

$$P(A \cap B) = P(A) \times P(B|A)$$




A common misconception is confusing $P(A|B)$ with $P(B|A)$. They are structurally different:

- $P(\text{Cough}|\text{Cold})$ is the probability that you have a cough, given you have a cold (which is very high).
- $P(\text{Cold}|\text{Cough})$ is the probability that you have a cold, given you have a cough (which is lower, as coughs can be caused by many other things).

### 12.1.2 Tree Diagrams

Tree diagrams are powerful tools for modelling sequential events. They help you map out all possible outcomes and calculate both intersection probabilities and total marginal probabilities.

- **Branches:** The first set of branches represents the **prior probabilities** of the first event. Subsequent branches represent **conditional probabilities**.
- **Multiplying along branches:** This gives the probability of the intersection of the events along that path ($P(A \cap B)$).
- **Summing paths:** To find the total probability of an event occurring at the end of the tree, you sum the probabilities of all paths that lead to that event.

**Example 1**

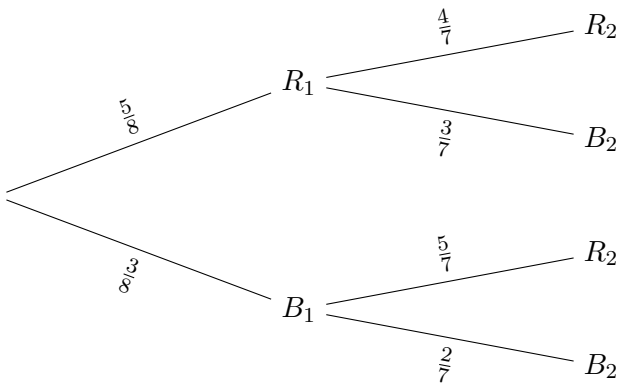
**Question:** A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn at random **without replacement**.

(a) Draw a tree diagram to represent this information.

(b) Calculate the probability that the second marble is red.

**Solution:**

(a) Let $R_1$ and $B_1$ be the events of drawing a red or blue marble first. Let $R_2$ and $B_2$ be the events for the second draw. Because the draw is without replacement, the probabilities on the second branches change.




(b) To find the total probability that the second marble is red, $P(R_2)$, we sum the probabilities of all paths ending in $R_2$:

$$P(R_2) = P(R_1 \cap R_2) + P(B_1 \cap R_2)$$

$$P(R_2) = \left(\frac{5}{8} \times \frac{4}{7}\right) + \left(\frac{3}{8} \times \frac{5}{7}\right)$$

$$P(R_2) = \frac{20}{56} + \frac{15}{56} = \frac{35}{56} = \frac{5}{8}$$

## Exercise 12A

**1** A biased coin has a probability of 0.6 of landing on Heads. It is flipped twice.

(a) Draw a tree diagram to represent the possible outcomes. **[2 marks]**

(b) Find the probability of getting exactly one Head. **[3 marks]**

**2** In a college, 60% of students study Maths. Of those who study Maths, 80% also study Physics. Of those who do not study Maths, 30% study Physics.

(a) Draw a tree diagram to model this scenario. **[2 marks]**

(b) Calculate the probability that a randomly selected student studies Physics. **[3 marks]**

(c) Given that a randomly selected student studies Physics, calculate the probability that they also study Maths. **[4 marks]**

**3** A factory produces components using two machines, $A$ and $B$. Machine $A$ produces 70% of the components and Machine $B$ produces the remaining 30%. It is known that 5% of the components produced by Machine $A$ are defective, while 2% of the components produced by Machine $B$ are defective.

(a) Draw a tree diagram to represent this information. **[2 marks]**

(b) Find the probability that a randomly selected component is defective. **[3 marks]**

(c) Given that a randomly selected component is defective, find the probability that it was produced by Machine $A$. **[4 marks]**

**4** In a particular city, it rains on 25% of days. When it rains, the probability that the morning train is delayed is 0.15. When it does not rain, the probability that the morning train is delayed is 0.04.

(a) Find the probability that the morning train is delayed on a randomly chosen day. **[3 marks]**

(b) Given that the morning train is delayed, find the probability that it was raining. **[3 marks]**

**5** A fair six-sided die is rolled. If the score is a multiple of 3, a token is drawn from Bag $A$. Otherwise, a token is drawn from Bag $B$. Bag $A$ contains 5 red tokens and 1 blue token. Bag $B$ contains $x$ red tokens and 2 blue tokens. Given that the drawn token is red, the probability that it came from Bag $A$ is $\frac{1}{3}$. Find the value of $x$. **[6 marks]**

**6** An IT support company receives requests via email (50%), phone (30%), and an online portal (20%). The probability that a request is resolved within an hour depends on the method of submission: 40% for email, 80% for phone, and $p$% for the online portal. A request is selected at random. Given that it was resolved within an hour, the probability that it was submitted via the online portal is 0.2. Find the value of $p$. **[5 marks]**

**7** At a wildlife reserve, 5% of the birds are of a rare protected species. An automated camera system uses image recognition to flag these rare birds. If a bird is of the rare species, the system correctly flags it 80% of the time. If a bird is not of the rare species, the system incorrectly flags it 10% of the time.

(a) Find the probability that a randomly observed bird is flagged by the system. **[3 marks]**

(b) Given that a bird is flagged, find the probability that it is actually of the rare species. **[3 marks]**

(c) The reserve managers want to upgrade the software so that if a bird is flagged, there is at least an 80% probability it is the rare species. Assuming the true positive rate remains 80% and the population proportion remains 5%, find the maximum acceptable false positive rate for the new software, giving your answer as an exact fraction. **[4 marks]**

**8** A magician has two biased coins in his pocket. Coin $A$ has a probability of 0.7 of landing on Heads. Coin $B$ has a probability of 0.4 of landing on Heads. He selects one coin at random and flips it three times.

(a) Find the probability that the coin lands on Heads exactly twice. **[4 marks]**

(b) Given that the coin landed on Heads exactly twice, find the probability that the magician selected Coin $A$. **[4 marks]**

**9** A local football team plays its home matches on a pitch that is heavily affected by the weather. The probability that the weather is Sunny, Cloudy, or Rainy on a match day is 0.40, 0.45, and 0.15 respectively. The probability that the team wins their match is 0.70 if it is Sunny, 0.50 if it is Cloudy, and 0.20 if it is Rainy.

(a) Find the probability that the team wins a randomly selected home match. **[3 marks]**

(b) Given that the team won their match, find the probability that the weather was Sunny. **[3 marks]**

(c) Given that the team did not win their match, find the probability that the weather was Rainy. **[4 marks]**

## 12.2 Bayes' Theorem

Often, we know the probability of an event $B$ given event $A$, $P(B|A)$, but we need to find the reverse: the probability of event $A$ given event $B$, $P(A|B)$. Bayes' theorem provides the mathematical framework to do this.



Figure 12.1: Thomas Bayes, Possibly (1701–1761)

### 12.2.1 Deriving Bayes' Theorem

From the conditional probability formula, we know:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Using the multiplication rule, we can replace the numerator $P(A \cap B)$ with $P(A) \times P(B|A)$:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

The denominator, $P(B)$, is the total probability of event $B$ occurring. Using the Law of Total Probability (as seen when summing paths on a tree diagram), $P(B) = P(A)P(B|A) + P(A')P(B|A')$.

**Key Point**

**Bayes' Theorem (for two events):**

$$P(A|B) = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A')P(B|A')}$$

- $P(A)$ is the **prior probability** (the initial probability of $A$ before observing $B$).
- $P(A|B)$ is the **posterior probability** (the updated probability of $A$ after observing $B$).

**Example 2**

**Question:** A rare disease affects 1 in 1000 people in a population. A diagnostic test for the disease is 99% accurate for those who have the disease (true positive rate) and 95% accurate for those who do not have the disease (true negative rate).
A person is chosen at random and tests positive. Calculate the probability that they actually have the disease.
**Solution:** Let $D$ be the event of having the disease, and $T$ be the event of testing positive. We are given:

- $P(D) = 0.001 \implies P(D') = 0.999$

- $P(T|D) = 0.99$
- $P(T'|D') = 0.95 \implies P(T|D') = 0.05$ (False positive rate)

We need to find $P(D|T)$. Using Bayes' theorem:

$$P(D|T) = \frac{P(D)P(T|D)}{P(T)}$$

First, calculate the total probability of a positive test, $P(T)$ (the denominator):

$$P(T) = P(D)P(T|D) + P(D')P(T|D')$$

$$P(T) = (0.001 \times 0.99) + (0.999 \times 0.05)$$

$$P(T) = 0.00099 + 0.04995 = 0.05094$$

Now, apply the formula:

$$P(D|T) = \frac{0.00099}{0.05094} \approx 0.0194 \text{ (3 s.f.)}$$

*Interpretation:* Even though the test is highly accurate, because the disease is so rare, a person who tests positive only has a 1.94% chance of actually having the disease. This is a classic example of the false positive paradox.

**Key Point**

**Common Exam Pitfalls:**

- **Prior vs. Posterior confusion:** The most frequent error is incorrectly identifying prior and posterior probabilities in complex word problems. Read the question carefully. Do not substitute $P(A|B)$ when the formula explicitly requires $P(B|A)$.
- **Denominator omission:** Remember that the denominator must include **all** valid paths that lead to the conditional event. Do not forget to add the false positive path!

## Exercise 12B

**1** Events $A$ and $B$ are such that $P(A) = 0.4$, $P(B|A) = 0.7$, and $P(B|A') = 0.2$.

(a) Calculate $P(B)$. **[2 marks]**

(b) Calculate $P(A|B)$. **[3 marks]**

**2** A factory has two machines, $A$ and $B$, which produce glass bottles. Machine $A$ produces 65% of the bottles and machine $B$ produces 35% of the bottles. The proportion of defective bottles produced by machine $A$ is 3% and the proportion of defective bottles produced by machine $B$ is 6%. A bottle is selected at random.

(a) Draw a tree diagram to represent this information. **[2 marks]**

(b) Find the probability that the bottle is defective. **[3 marks]**

(c) Given that the bottle is defective, find the probability that it was produced by machine $B$. **[4 marks]**

**3** Events $C$ and $D$ are such that $P(D|C) = 0.8$ and $P(D|C') = 0.3$. The probability of event $C$ occurring is $p$.

(a) Find an expression for $P(D)$ in terms of $p$. **[2 marks]**

(b) Given that $P(C|D) = 0.6$, find the exact value of $p$. **[4 marks]**

**4** A spam filter identifies 90% of spam emails correctly. However, it also incorrectly flags 5% of legitimate emails as spam. It is known that 20% of all emails received are spam. If an email is flagged as spam, what is the probability that it is actually a legitimate email? **[4 marks]**

**5** In a factory, Machine X produces 60% of the components and Machine Y produces 40%. The defect rate for Machine X is 3% and for Machine Y is 5%. A component is chosen at random and found to be defective. Find the probability it was produced by Machine Y. **[4 marks]**

## 12.3 Bayes' Theorem for Three Events

Bayes' theorem can be extended to scenarios involving more than two mutually exclusive starting events. The Edexcel specification requires you to apply Bayes' theorem for up to three events.

**Key Point**

**General Formula for Bayes' Theorem:**

$$P(A_j|B) = \frac{P(A_j)P(B|A_j)}{\sum P(A_i)P(B|A_i)}$$

For three mutually exclusive and exhaustive events $A_1, A_2, A_3$, the denominator expands to:

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

**Example 3**

**Question:** A company sources microchips from three different suppliers: Alpha, Beta, and Gamma.

- Alpha supplies 50% of the chips.
- Beta supplies 30% of the chips.
- Gamma supplies 20% of the chips.

The probability that a chip is defective depends on the supplier: 1% for Alpha, 2% for Beta, and 4% for Gamma.
A chip is selected at random from the company's inventory and is found to be defective. Calculate the probability that it was supplied by Beta.
**Solution:** Let $A, B, G$ represent the suppliers Alpha, Beta, and Gamma. Let $D$ represent a defective chip. We are given:

- Priors: $P(A) = 0.50$, $P(B) = 0.30$, $P(G) = 0.20$
- Likelihoods: $P(D|A) = 0.01$, $P(D|B) = 0.02$, $P(D|G) = 0.04$

We need to find $P(B|D)$. First, calculate the total probability of a defective chip, $P(D)$ (the denominator):

$$P(D) = P(A)P(D|A) + P(B)P(D|B) + P(G)P(D|G)$$

$$P(D) = (0.50 \times 0.01) + (0.30 \times 0.02) + (0.20 \times 0.04)$$

$$P(D) = 0.005 + 0.006 + 0.008 = 0.019$$

Now, apply Bayes' theorem:

$$P(B|D) = \frac{P(B)P(D|B)}{P(D)}$$

$$P(B|D) = \frac{0.006}{0.019} = \frac{6}{19} \approx 0.316 \text{ (3 s.f.)}$$

### Exercise 12C

**1** A certain genetic marker is present in 4% of a population. A diagnostic test for this marker is 95% sensitive (it correctly identifies the presence of the marker 95% of the time) and 98% specific (it correctly identifies the absence of the marker 98% of the time).

(a) A randomly selected person from the population tests positive. Calculate the probability that they actually have the genetic marker. **[4 marks]**

(b) Explain why this probability might be lower than a patient would expect, despite the high sensitivity and specificity of the test. **[1 mark]**

**2** A bakery produces croissants using three different ovens: A, B, and C. Oven A produces 40% of the croissants, Oven B produces 35%, and Oven C produces 25%. The probability that a croissant is burnt is 0.03 for Oven A, 0.05 for Oven B, and 0.02 for Oven C.

(a) Find the probability that a randomly selected croissant is burnt. **[2 marks]**

(b) Given that a randomly selected croissant is burnt, find the probability it was baked in Oven B. **[3 marks]**

**3** A box contains 7 dark chocolates and 5 milk chocolates. Two chocolates are selected at random without replacement.

(a) Draw a tree diagram to represent this information. **[2 marks]**

(b) Find the probability that both chocolates are the same type. **[2 marks]**

(c) Given that the second chocolate selected is dark, find the probability that the first chocolate selected was milk. **[3 marks]**

**4** A survey of 200 university students recorded their primary study location and whether they achieved a first-class grade.

- 80 students primarily study in the Library.
- 70 students primarily study in their Room.
- 50 students primarily study in a Cafe.

The proportion of students achieving a first-class grade is 30% for those studying in the Library, 20% for those in their Room, and 16% for those in a Cafe.

(a) A student is chosen at random. Calculate the probability they achieved a first-class grade. **[2 marks]**

(b) Given that a chosen student did not achieve a first-class grade, find the probability they primarily study in their Room. **[4 marks]**

**5** An email filter classifies incoming emails as either 'Spam' or 'Safe'. Historically, 45% of all incoming emails are Spam. The filter correctly identifies 92% of Spam emails as 'Spam'. However, it also incorrectly classifies a certain proportion, $y$, of Safe emails as 'Spam'. Given that an email is classified as 'Spam' by the filter, the probability that it is actually Safe is 0.05.

(a) Find the value of $y$, giving your answer to 3 significant figures. **[4 marks]**

(b) Find the overall probability that an email is correctly classified by the filter. **[3 marks]**

**6** A weather forecasting model predicts tomorrow's weather as Sunny, Cloudy, or Rainy. The model predicts Sunny with probability 0.5, Cloudy with probability 0.3, and Rainy with probability 0.2. If the model predicts Sunny, it actually rains 5% of the time. If the model predicts Cloudy, it actually rains 25% of the time. If the model predicts Rainy, it actually rains 80% of the time.

(a) Find the probability that it rains tomorrow. **[2 marks]**

(b) Given that it does not rain tomorrow, find the probability that the model predicted it would be Sunny. **[4 marks]**

## 12.4 Solving Problems with Unknowns

In more advanced exam questions, you may be given the final posterior probability $P(A|B)$ alongside the branch probabilities, but one prior probability or conditional probability is left as an unknown variable (e.g., '$x$'). You must set up Bayes' theorem algebraically and solve for the unknown.

**Example 4**

**Question:** There are two bags of counters. Bag 1 contains 3 red counters and 2 blue counters. Bag 2 contains $x$ red counters and 4 blue counters.
A fair coin is flipped. If it lands on Heads, a counter is drawn from Bag 1. If it lands on Tails, a counter is drawn from Bag 2. Given that the drawn counter is red, the probability that it came from Bag 1 is $\frac{9}{14}$.
Find the value of $x$.

**Solution:** Let $B_1$ and $B_2$ be the events of choosing Bag 1 and Bag 2. Let $R$ be the event of drawing a red counter. We are given:

- $P(B_1) = 0.5$, $P(B_2) = 0.5$
- $P(R|B_1) = \frac{3}{5} = 0.6$
- $P(R|B_2) = \frac{x}{x+4}$
- $P(B_1|R) = \frac{9}{14}$

Set up Bayes' theorem for $P(B_1|R)$:

$$P(B_1|R) = \frac{P(B_1)P(R|B_1)}{P(B_1)P(R|B_1) + P(B_2)P(R|B_2)}$$

Substitute the known values:

$$\frac{9}{14} = \frac{0.5 \times 0.6}{(0.5 \times 0.6) + \left(0.5 \times \frac{x}{x+4}\right)}$$

Simplify the numerator and denominator:

$$\frac{9}{14} = \frac{0.3}{0.3 + 0.5\left(\frac{x}{x+4}\right)}$$

Cross-multiply to solve for $x$:

$$9\left(0.3 + 0.5\left(\frac{x}{x+4}\right)\right) = 14 \times 0.3$$

$$2.7 + 4.5\left(\frac{x}{x+4}\right) = 4.2$$

Subtract 2.7 from both sides:

$$4.5\left(\frac{x}{x+4}\right) = 1.5$$

Divide by 4.5:

$$\frac{x}{x+4} = \frac{1.5}{4.5} = \frac{1}{3}$$

Cross-multiply again:

$$3x = x + 4$$

$$2x = 4$$

$$x = 2$$

So, there are 2 red counters in Bag 2.

### 12.4.1 Real-World Applications and the Base Rate Fallacy

Bayes' theorem is incredibly important in medical testing and real-world diagnostics. A common error made by the general public, the media, and even medical professionals is ignoring the underlying probability of a condition (the prior probability or "base rate") when interpreting test results. This mathematical blind spot is known as the **Base Rate Fallacy**.

**Example 5**

**Question:** During the COVID-19 pandemic, Lateral Flow Tests (LFTs) were widely used. Suppose a specific LFT has a **sensitivity of 80%** (if a person has the virus, the test is positive 80% of the time) and a **specificity of 99%** (if a person does not have the virus, the test is negative 99% of the time, meaning there is a 1% false positive rate).
Assume the test is given to a randomly selected person from the general public during a period of low prevalence, where only 1% of the population actually has the virus.
If the person tests positive, calculate the probability that they actually have COVID-19.
**Solution:** Let $C$ be the event that the person has COVID-19. Let $T^+$ be the event that the test result is positive.
We are given the following probabilities:

- **Prior probabilities (Base rate):** $P(C) = 0.01 \implies P(C') = 0.99$
- **Conditional probabilities (Test accuracy):**
  - True positive rate: $P(T^+|C) = 0.80$
  - False positive rate: $P(T^+|C') = 1 - 0.99 = 0.01$

We need to find the posterior probability, $P(C|T^+)$.
Set up Bayes' theorem:

$$P(C|T^+) = \frac{P(C)P(T^+|C)}{P(C)P(T^+|C) + P(C')P(T^+|C')}$$

Substitute the known values:

$$P(C|T^+) = \frac{0.01 \times 0.80}{(0.01 \times 0.80) + (0.99 \times 0.01)}$$

Simplify the numerator and the denominator:

$$P(C|T^+) = \frac{0.008}{0.008 + 0.0099}$$

$$P(C|T^+) = \frac{0.008}{0.0179} \approx 0.447 \text{ (3 s.f.)}$$

**Interpretation:** Even though the test is 99% specific and 80% sensitive, a person with a positive result only has a **44.7%** chance of actually having the virus. More than half of all positive tests in this scenario are false positives!
How is this mathematically possible? Because the virus is so rare (1 in 100 people), the sheer volume of healthy people taking the test means that the tiny 1% error rate generates more false positives (0.0099) than there are actual sick people to generate true positives (0.008).

### Exercise 12D

**1** A test for a virus has a true positive rate of 98% ($P(\text{Positive}|\text{Virus}) = 0.98$). The false positive rate is $y$ ($P(\text{Positive}|\text{No Virus}) = y$). The virus is present in 5% of the population. Given that a person who tests positive has a 70% chance of actually having the virus, find the false positive rate $y$. **[5 marks]**

**2** Two archers, Alice and Bob, shoot at a target. Alice shoots 60% of the arrows and Bob shoots 40%. Alice hits the bullseye with probability $p$. Bob hits the bullseye with probability 0.3. Given that an arrow hitting the bullseye was shot by Alice with probability $\frac{2}{3}$, find the value of $p$. **[4 marks]**

**3** A factory has three production lines: A, B, and C. Line A produces 40% of the items, Line B produces 35%, and Line C produces 25%. The defect rates are 2% for A, 4% for B, and $k$% for C. If a randomly selected defective item has a probability of $\frac{1}{4}$ of coming from Line C, find the value of $k$. **[4 marks]**

**4** A spam filter correctly identifies 96% of spam emails. However, it also incorrectly flags 1% of legitimate emails as spam. Given that an email is flagged as spam by the system, the probability that it is actually a spam email is 0.96. Find the overall percentage of emails received by the system that are spam. **[4 marks]**

**5** A commuter travels to work by Bus, Train, or Car. She is equally likely to take the Bus or the Train. The probability she is late when taking the Bus is 0.2, when taking the Train is 0.1, and when taking the Car is 0.05. Given that the commuter is late, the probability she took the Bus is 0.5. Find the probability that she takes the Car. **[4 marks]**

**6** A bag contains 10 coins. $n$ of these coins are biased such that the probability of landing on Heads is 0.8. The remaining $10 - n$ coins are standard fair coins. A coin is drawn at random from the bag and flipped once. It lands on Heads. Given that the coin landed on Heads, the probability that it was one of the biased coins is $\frac{8}{13}$. Set up an algebraic equation using Bayes' theorem and solve it to find the value of $n$. **[5 marks]**

### Beyond the Syllabus: The Great Debate: Bayesian vs. Frequentist Statistics

The inferential methods in A-Level Statistics (hypothesis testing, $p$-values, confidence intervals) are built on one specific philosophy of probability: **Frequentist Statistics**. But Bayes' Theorem opens the door to a rival philosophy that is now widely used in data science and machine learning: **Bayesian Statistics**.
The debate between the two camps is one of the longest-running in the history of statistics. It

comes down to a fundamental disagreement about what "probability" actually means.

**1. The Frequentist View (The A-Level Way)**

To a Frequentist, probability is the *long-run frequency* of an event. If you flip a fair coin more and more times, the proportion of heads settles towards 0.5. Furthermore, Frequentists believe that **population parameters are fixed, but data is random**. For example, the true mean weight of all apples in an orchard ($\mu$) is a fixed, unchanging number. You just don't know it. Because $\mu$ is fixed, a Frequentist will never say "There is a 95% probability that $\mu$ is 120g." Instead, they say, "If we repeated the sampling many times, about 95% of the confidence intervals constructed in this way would contain the fixed $\mu$."

**2. The Bayesian View (The Degree-of-Belief Way)**

To a Bayesian, probability is a *degree of belief* or certainty. Bayesians believe that **data is fixed, but parameters are random**. You have your observed data. It is sitting right in front of you, so it's not random. The unknown parameter ($\mu$) is what you are uncertain about, so Bayesians treat $\mu$ as a random variable with its own probability distribution!

A Bayesian starts with a **Prior Belief** (e.g., "I think the apple weighs 120g, but I'm not sure"). Then, they collect data. Finally, they use Bayes' Theorem to mathematically update their belief, creating a **Posterior Belief**. As more and more data comes in, the Bayesian continuously updates their distribution, which typically concentrates more and more tightly around the true value.

Today, Bayesian updating is used throughout machine learning and engineering. A naive Bayes spam filter updates the probability that an email is spam as it learns from new messages, and the tracking filters used in self-driving cars update their estimate of where a pedestrian will walk in exactly this way.

## End-of-Chapter Mixed Exercises

1. A company manufactures computer chips using two machines, 'Machine A' and 'Machine B'. Machine A produces 60% of the chips and Machine B produces the remaining 40%. It is known that 3% of the chips produced by Machine A are defective, while 5% of the chips produced by Machine B are defective. A chip is selected at random from the total production.

   (a) Calculate the probability that the selected chip is defective. **[3 marks]**

   (b) Given that the selected chip is defective, use Bayes' theorem to find the probability that it was produced by Machine A. **[3 marks]**

2. A specific genetic mutation occurs in 1 in 500 people within a population. A diagnostic test has been developed to detect this mutation. If a person has the mutation, the test will correctly identify it with a probability of 0.99. If a person does not have the mutation, the test will incorrectly return a positive result with a probability of 0.02.

   (a) Draw a tree diagram to represent this information. **[3 marks]**

   (b) Find the probability that a randomly selected person tests positive for the mutation. **[3 marks]**

   (c) A person is chosen at random and tests positive. Calculate the probability that they actually possess the genetic mutation. **[4 marks]**

   (d) Briefly explain why the answer to part (c) is so low, despite the test being highly accurate. **[1 mark]**

3. Students at a large sixth form college commute by bus, train, or car. The probabilities of a student using each method of transport are 0.45, 0.35, and 0.20 respectively. The probability that a student is late for college given that they travel by bus is 0.12. The corresponding probabilities for train and car are 0.08 and 0.05 respectively.

   (a) Find the probability that a randomly chosen student is late for college. **[3 marks]**

   (b) Given that a student is late, find the probability that they travelled by train. **[3 marks]**

   (c) Given that a randomly chosen student is "on time", find the probability that they travelled by car. **[4 marks]**

4. A biased coin has a probability $p$ of landing on heads. The coin is tossed once. If it lands on heads, a standard fair six-sided die is rolled. If it lands on tails, a biased six-sided die is rolled, for which the probability of rolling a six is $\frac{1}{3}$. The overall probability of rolling a six is $\frac{1}{4}$.

   (a) Show that $p = \frac{1}{2}$. **[4 marks]**

   (b) Given that a six was rolled, find the probability that the coin landed on heads. **[3 marks]**

5. A bag contains 5 indistinguishable coins. Three of the coins are fair, one coin is biased such that the probability of landing on heads is 0.75, and the final coin is double-headed (the probability of landing on heads is 1). A coin is selected at random from the bag and tossed three times.

   (a) Calculate the probability that all three tosses result in heads. **[4 marks]**

   (b) Given that the first three tosses all resulted in heads, find the probability that the double-headed coin was selected. **[4 marks]**

   (c) The same coin is now tossed a fourth time. Given that the first three tosses were heads, find the exact probability that the fourth toss is also a head. **[4 marks]**

6. A certain medical disease affects 1 in every 2500 people in the population. A screening test for this disease gives a positive result for 98% of people who have the disease. It also gives a false positive result for 1% of people who do not have the disease.

   (a) Find the probability that a person chosen at random tests positive for the disease. **[3 marks]**

   (b) Given that a randomly chosen person tests positive, find the probability that they actually have the disease. Give your answer to 3 significant figures. **[4 marks]**

   (c) Comment on the practicality of using this screening test for the general population. **[1 mark]**

7. In a factory, three machines, $X$, $Y$, and $Z$, produce respectively 20%, 35%, and 45% of the total output. Past records show that 4%, 2%, and 1% of the items produced by machines $X$, $Y$, and $Z$ respectively are defective. A quality control inspector selects an item at random from the total output.

   (a) Find the probability that the item selected is defective. **[3 marks]**

   (b) Given that the selected item is defective, use Bayes' theorem to calculate the probability that it was produced by machine $Y$. **[4 marks]**

8. A jar contains 6 orange sweets and 4 lime sweets. Two sweets are selected at random without replacement.

   (a) Draw a tree diagram to represent the two selections. **[2 marks]**

   (b) Find the probability that both sweets selected are the same flavour. **[3 marks]**

9. An airport security scanner is designed to detect prohibited items hidden in luggage. The scanner correctly flags 95% of bags that contain a prohibited item, but it also incorrectly flags 3% of bags that do not contain one. Only 0.2% of bags passing through the scanner actually contain a prohibited item.

   (a) Find the probability that a randomly selected bag is flagged by the scanner. **[3 marks]**

   (b) Given that a bag is flagged, find the probability that it actually contains a prohibited item. Give your answer to 3 significant figures. **[4 marks]**

   (c) Briefly explain why this probability is so low despite the scanner's high accuracy. **[1 mark]**

10. A courier company routes parcels through three regional depots: North, South, and East. The North depot handles 35% of parcels, the South depot handles 40%, and the East depot handles 25%. The probability that a parcel is delivered late is 6% for the North depot, 3% for the South depot, and 9% for the East depot.

    (a) Find the probability that a randomly selected parcel is delivered late. **[3 marks]**

    (b) Given that a parcel was delivered late, find the probability that it was routed through the North depot. **[3 marks]**

    (c) Given that a parcel was delivered on time, find the probability that it was routed through the East depot. **[4 marks]**

11. In a laboratory culture, a proportion $q$ of bacterial colonies belong to Strain P, and the remainder belong to Strain Q. The probability that a colony develops antibiotic resistance is 0.6 for Strain P and 0.15 for Strain Q. Given that a randomly selected resistant colony has probability $\frac{5}{7}$ of belonging to Strain P, find the exact value of $q$. **[5 marks]**

12. A drawer contains 5 blue socks and 3 grey socks. Socks are removed one at a time at random, without replacement.

(a) Draw a tree diagram to show the outcomes of the first two socks removed. **[2 marks]**

(b) Find the probability that the second sock removed is blue. **[3 marks]**

13. A car dealership sells both new and used cars. 55% of sales are new cars, and the remainder are used cars. The probability that a customer requests an extended service warranty is 0.8 for a new car sale and 0.35 for a used car sale.

(a) Find the probability that a randomly selected sale includes a request for an extended warranty. **[3 marks]**

(b) Given that a sale included a warranty request, find the probability that it was a used car. Give your answer to 3 significant figures. **[3 marks]**

# Chapter 12 Summary

**Summary of Key Formulas**

1. **Conditional Probability:**
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

2. **Multiplication Rule:**
$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

3. **Law of Total Probability (Denominator of Bayes' Theorem):** For mutually exclusive and exhaustive events $A_1, A_2, \ldots, A_n$:
$$P(B) = \sum_{i=1}^{n} P(A_i)P(B|A_i)$$

4. **Bayes' Theorem:**
$$P(A_j|B) = \frac{P(A_j)P(B|A_j)}{\sum P(A_i)P(B|A_i)}$$

5. **Terminology:**
   - $P(A_j)$ is the **prior probability**.
   - $P(A_j|B)$ is the **posterior probability**.

# Chapter 17

# Hypothesis Testing for 1 and 2 Samples

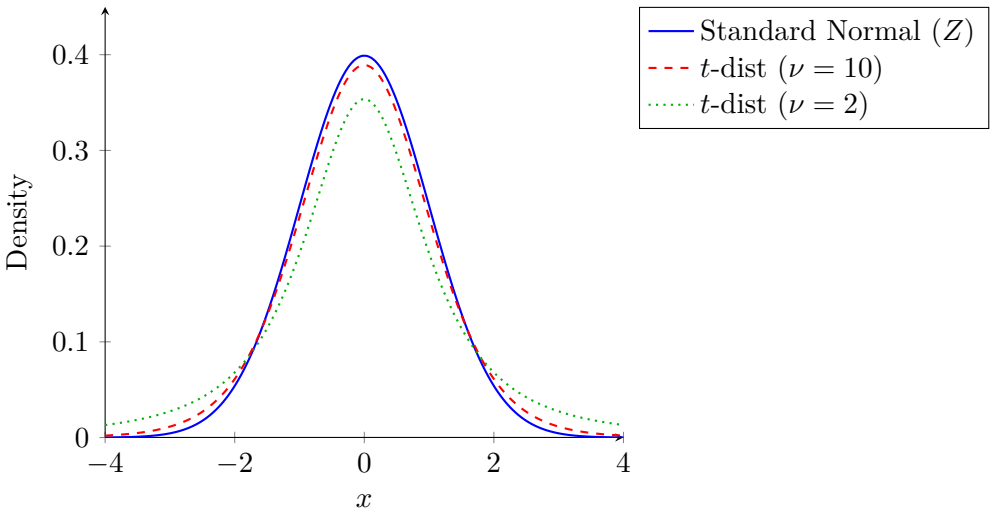
**After completing this chapter you should be able to:**

- Conduct tests for the mean of a normal distribution with unknown variance using the t-distribution.
- Conduct tests for the difference of two means for two independent normal distributions with known variances.
- Conduct tests for the difference of two means for two independent normal distributions with unknown but equal variances.
- Calculate and interpret confidence intervals for the difference between two means.
- Conduct large sample tests for the difference of two means when population variances are unknown.
- Conduct tests for the difference between two binomial proportions.
- Interpret results for these tests in context.

## 17.1 1-Sample $t$-tests

In previous chapters, you learned how to conduct a hypothesis test for the mean of a normal distribution when the population variance ($\sigma^2$) is known, using the standard normal distribution ($Z$-distribution). However, in practice, the population variance is rarely known.

When $\sigma^2$ is unknown, we must estimate it using the unbiased sample variance, $S^2$. Because we are using an estimate for the variance, the test statistic no longer follows a normal distribution; instead, it follows a Student's $t$-distribution. The shape of the $t$-distribution depends on the sample size, specifically the **degrees of freedom** ($\nu$).




**Key Point**

For a random sample of size $n$ taken from a normal distribution $N(\mu, \sigma^2)$ where $\sigma^2$ is unknown:

- **1-sample $t$-test statistic:**

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

- $S$ is the unbiased sample standard deviation.
- The degrees of freedom is $\nu = n - 1$.

If you are given raw data or summary statistics ($\sum x$ and $\sum x^2$), you can calculate the unbiased sample variance using the formula:

$$S^2 = \frac{1}{n-1}\sum(x - \bar{x})^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)$$

Most modern calculators will compute $S$ directly in Statistics mode (often denoted as $sx$ or $\sigma_{n-1}$).

**Example 1**

**Question:** A machine fills bags with flour. The weight of flour in a bag is assumed to be normally distributed. The manufacturer claims the mean weight is 500g. A random sample of 8 bags is taken and their weights, in grams, are:

$$495, \quad 502, \quad 498, \quad 491, \quad 505, \quad 496, \quad 492, \quad 499$$

Test, at the 5% significance level, whether there is evidence that the mean weight is less than 500g.
**Solution:**

- **Hypotheses:** Let $\mu$ be the population mean weight of the bags.

$$H_0 : \mu = 500$$

$$H_1 : \mu < 500$$

- **Summary Statistics:** Using a calculator on the raw data:

$$n = 8, \quad \bar{x} = 497.25, \quad S = 4.7734...$$

- **Test Statistic:**
$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{497.25 - 500}{4.7734/\sqrt{8}} = \frac{-2.75}{1.6877} = -1.629$$
- **Critical Value:** Degrees of freedom $\nu = 8 - 1 = 7$. For a 5% one-tailed test, we look up the critical value in the $t$-distribution tables: $t_7(0.05) = -1.895$.
- **Conclusion:** Since $-1.629 > -1.895$, the test statistic does not fall in the critical region. We do not reject $H_0$. There is insufficient evidence at the 5% level to suggest the mean weight of the bags is less than 500g.

## Exercise 17A

**1** A sample of size 15 is taken from a normal distribution with unknown variance. The sample mean is 102.5 and the unbiased sample standard deviation is 8.4.

(a) State the degrees of freedom for a 1-sample $t$-test. **[1 mark]**

(b) Calculate the test statistic to test $H_0 : \mu = 100$. **[2 marks]**

**2** A local bakery claims that their artisan loaves of bread have a mean weight of 800 g. A trading standards officer suspects that the loaves are underweight. She takes a random sample of 7 loaves and records their weights, in grams, as follows:

$$791, \quad 785, \quad 802, \quad 794, \quad 788, \quad 796, \quad 789$$

(a) State the assumption necessary to carry out a 1-sample $t$-test. **[1 mark]**

(b) Test, at the 5% significance level, whether there is evidence to support the officer's suspicion. **[6 marks]**

**3** An environmental scientist is measuring the concentration of a specific pollutant in a river. The concentration is assumed to be normally distributed. The historical mean concentration is known to be 3.5 mg/L. A random sample of 20 water specimens is collected. The sample mean is 3.82 mg/L and the unbiased sample variance is 0.45 $(\text{mg/L})^2$.

(a) Test, at the 10% significance level, whether the mean concentration of the pollutant has changed. State your hypotheses clearly. **[6 marks]**

(b) Explain why a $t$-test was used instead of a $z$-test. **[1 mark]**

**4** A car manufacturer claims that a new model of electric vehicle has a mean range of 250 miles on a full charge. An independent reviewer believes the actual mean range is less than this. The reviewer tests a random sample of 14 vehicles and records their ranges, $x$ miles. The results are summarised by:

$$\sum(x - 250) = -42, \quad \sum(x - 250)^2 = 850$$

Assume that the range of the vehicles is normally distributed.

(a) Calculate the sample mean and the unbiased sample standard deviation. **[3 marks]**

(b) Conduct a hypothesis test at the 1% significance level to assess the reviewer's belief. **[5 marks]**

**5** A factory produces steel cables. The breaking strength of the cables is normally distributed. The factory claims that the mean breaking strength is at least 40 kN. A quality control inspector takes a random sample of 9 cables and finds a sample mean of 38.5 kN and an unbiased sample standard deviation of $s$ kN.

The inspector conducts a 1-sample $t$-test at the 5% significance level and finds that the result is significant, leading to the rejection of the factory's claim.

Find the upper boundary for possible values of $s$, giving your answer to 2 decimal places. **[4 marks]**

**6** A software company claims that their new compression algorithm reduces file sizes to a mean of 45 MB. A user tests the algorithm on a random sample of 25 files. The compressed file sizes, $y$ MB, are summarised by:

$$\sum y = 1165, \quad \sum y^2 = 54820$$

(a) Calculate the unbiased sample variance of the compressed file sizes. **[2 marks]**

(b) Test, at the 2% significance level, whether the mean compressed file size differs from 45 MB. **[5 marks]**

(c) The user discovers that the compressed file sizes were not normally distributed. State, giving a reason, whether your conclusion in part (b) is still valid. **[2 marks]**

**7** A barista claims that their new espresso machine dispenses a mean volume of 30 ml per shot. The shop manager suspects the machine is dispensing a different amount and decides to test this claim. The manager measures the volume, in ml, of a random sample of 8 espresso shots. The results are:

$$29.5, \quad 31.2, \quad 28.8, \quad 30.5, \quad 29.1, \quad 28.5, \quad 29.9, \quad 29.3$$

(a) State the assumption that must be made about the volume of espresso shots to carry out a 1-sample $t$-test. **[1 mark]**

(b) Test, at the 10% significance level, whether there is evidence to support the manager's suspicion. **[6 marks]**

## 17.2 2-Sample Tests for Means

Often, we want to compare the means of two different populations, such as the effectiveness of two different drugs or the yields of two different crop varieties. The test we choose depends on whether the population variances are known or unknown.

### 17.2.1 2-Sample $z$-test (Known Variances)

If we are comparing two independent normal distributions and both population variances ($\sigma_x^2$ and $\sigma_y^2$) are explicitly known, we use a 2-sample $z$-test.

**Key Point**

For two independent normal distributions with known variances $\sigma_x^2$ and $\sigma_y^2$:

- **2-sample $z$-test statistic:**

$$z = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}} \sim N(0, 1)$$

Usually, the null hypothesis is $H_0 : \mu_x = \mu_y$, which means $\mu_x - \mu_y = 0$.

**Example 2**

**Question:** The lifespans of two brands of lightbulbs, Brand A and Brand B, are normally distributed. The population variance for Brand A is known to be 1200 hours$^2$ and for Brand B is 900 hours$^2$. A random sample of 40 Brand A bulbs has a mean lifespan of 1050 hours. A random sample of 50 Brand B bulbs has a mean lifespan of 1020 hours. Test at the 1% significance level whether the mean lifespan of Brand A is greater than Brand B.
**Solution:**

- **Hypotheses:**
$$H_0 : \mu_A = \mu_B \quad (\text{or } \mu_A - \mu_B = 0)$$
$$H_1 : \mu_A > \mu_B \quad (\text{or } \mu_A - \mu_B > 0)$$
- **Test Statistic:** Since population variances are known, we use the $z$-test.
$$z = \frac{(\bar{x}_A - \bar{x}_B) - 0}{\sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}} = \frac{1050 - 1020}{\sqrt{\frac{1200}{40} + \frac{900}{50}}} = \frac{30}{\sqrt{30 + 18}} = \frac{30}{\sqrt{48}} \approx 4.330$$
- **Critical Value:** For a 1% one-tailed test, the critical $z$-value is 2.326.
- **Conclusion:** Since $4.330 > 2.326$, we reject $H_0$. There is sufficient evidence at the 1% level to conclude that Brand A lightbulbs have a longer mean lifespan than Brand B.

### 17.2.2 2-Sample $z$-test (Large Samples, Unknown Variances)

If the population variances ($\sigma_x^2$ and $\sigma_y^2$) are **unknown**, but **both sample sizes are large** ($n_x \geq 30$ and $n_y \geq 30$), we can use the Central Limit Theorem. We estimate the population variances using the unbiased sample variances ($S_x^2$ and $S_y^2$) and perform a $z$-test. We do **not** need to pool the variances, and we do not need to assume the population variances are equal.

**Key Point**

For two independent large samples with unknown variances:

- **Large Sample 2-sample $z$-test statistic:**
$$z = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}}} \sim N(0, 1)$$

### 17.2.3 2-Sample $t$-test (Unknown but Equal Variances)

If the population variances are unknown, we must estimate them from our samples. If we can assume that the two populations have the **same** unknown variance ($\sigma_x^2 = \sigma_y^2 = \sigma^2$), we can combine our two sample variances to get a better overall estimate of this common variance. This is called the **pooled variance**, $S_p^2$.

### Key Point

For two independent normal distributions with unknown but equal variances:

- **Pooled variance:**
$$S_p^2 = \frac{(n_x - 1)S_x^2 + (n_y - 1)S_y^2}{n_x + n_y - 2}$$
- **2-sample $t$-test statistic (equal variances):**
$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{S_p^2\left(\frac{1}{n_x} + \frac{1}{n_y}\right)}} \sim t_{n_x+n_y-2}$$
- The degrees of freedom is $\nu = n_x + n_y - 2$.

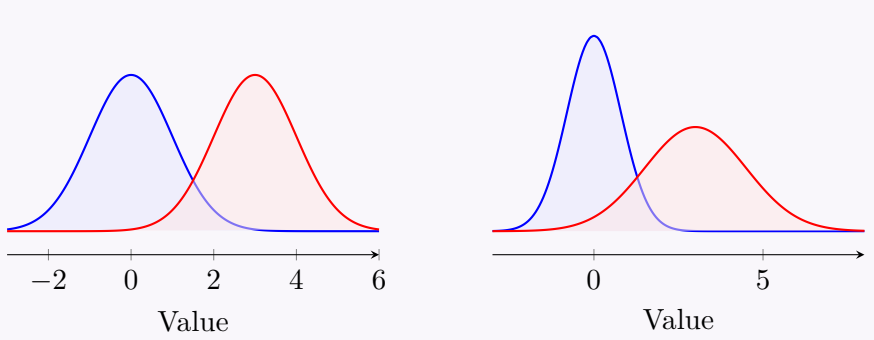
### Exam Pitfalls and Misconceptions

- **Testing for equal variances:** Do not waste time attempting to mathematically prove that the two population variances are equal before conducting a 2-sample $t$-test. The syllabus explicitly states candidates are not expected to test for equality of variances; it will be assumed or stated in the question.

  *However, this is required for Further Statistics 2 in the Edexcel Further Mathematics specification - see Section 4.*
- **Degrees of freedom errors:** Remember to use the combined degrees of freedom ($n_x+n_y-2$) for the 2-sample $t$-test. Do not accidentally use it for a 2-sample $z$-test (which uses the standard normal distribution).
- **Population vs. Sample variance:** When population variances are unknown, use the method specified for the setting. For small normal samples this is a $t$-test; for the large-sample approximation described in Section 17.2, sample variances are substituted in the large-sample $z$ statistic.

### Beyond the Syllabus: Welch's $t$-test: When Variances Aren't Equal

In your syllabus, you are told to assume that the two populations have equal variances ($\sigma_x^2 = \sigma_y^2$). This allows you to calculate a **pooled variance** ($S_p^2$). But what if the variances are clearly different?
For example, a new experimental teaching method might raise the average score, but it might also drastically increase the spread of the scores (some students thrive, others get left behind). If the true variances are unequal (a condition called **heteroscedasticity**) and your sample sizes are different, the standard Student's $t$-test can become unreliable: when the smaller sample comes from the more variable population, the chance of a Type I error can be far higher than the stated significance level.




**The Modern Solution: Welch's *t*-test**

For this reason, many statisticians now recommend **Welch's *t*-test** as the default choice. Welch's test doesn't bother pooling the variances. Instead, it relies on the rule you learned for combining independent random variables: $\text{Var}(\bar{X} - \bar{Y}) = \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) = \frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}$. Because we don't know the population variances, we just substitute the individual unbiased sample variances ($S_x^2$ and $S_y^2$). The test statistic looks almost identical to the 2-sample $z$-test, but uses sample quantities instead of population parameters:

$$t = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{\sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}}}$$

**The Magic of Fractional Degrees of Freedom**

The real genius of Welch's test is how it calculates the degrees of freedom ($\nu$). Because we didn't pool the variances, the standard $\nu = n_x + n_y - 2$ no longer works. Instead, we use the terrifying-looking **Welch-Satterthwaite equation**:

$$\nu = \frac{\left(\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}\right)^2}{\frac{(S_x^2/n_x)^2}{n_x - 1} + \frac{(S_y^2/n_y)^2}{n_y - 1}}$$

This formula adjusts the degrees of freedom downward based on how different the variances and sample sizes are. The result is often a *decimal* (e.g., $\nu = 17.4$). While you can't easily look up a decimal degree of freedom in a printed A-Level table, statistical software handles it effortlessly. In fact, R's `t.test` function performs Welch's $t$-test by default; you have to request the pooled Student's $t$-test that you learn in A-Level explicitly. (Python's SciPy works the other way round: `ttest_ind` pools the variances unless you set `equal_var=False`.)

**Example 3**

**Question:** A teacher wishes to compare the marks of students taught by two different methods. The marks are assumed to be normally distributed with equal population variances.

| Method | Sample Size ($n$) | Sample Mean ($\bar{x}$) | Sample Variance ($S^2$) |
|---|---|---|---|
| 1 | 10 | 65.2 | 15.4 |
| 2 | 12 | 60.5 | 18.2 |

Test at the 5% significance level whether there is a difference in the mean marks.
**Solution:**

- **Hypotheses:**

$$H_0 : \mu_1 = \mu_2$$
$$H_1 : \mu_1 \neq \mu_2$$

- **Pooled Variance:**

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} = \frac{9(15.4)+11(18.2)}{10+12-2} = \frac{138.6+200.2}{20} = 16.94$$

- **Test Statistic:**

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{65.2-60.5}{\sqrt{16.94\left(\frac{1}{10}+\frac{1}{12}\right)}} = \frac{4.7}{\sqrt{16.94 \times 0.1833...}} = \frac{4.7}{1.7623...} \approx 2.667$$

- **Critical Value:** Degrees of freedom $\nu = 10+12-2 = 20$. For a 5% two-tailed test, the critical value is $t_{20}(0.025) = 2.086$.
- **Conclusion:** Since $2.667 > 2.086$, we reject $H_0$. There is sufficient evidence at the 5% level to suggest a difference in the mean marks between the two teaching methods.

### 17.2.4 Confidence Intervals for the Difference Between Two Means

Just as we can perform hypothesis tests for the difference between two means, we can construct confidence intervals to estimate the true difference $(\mu_x - \mu_y)$.

**Key Point**

**Confidence Interval (Equal but Unknown Variances):**
If we assume the two normal populations have equal variances, we use the pooled variance $S_p^2$ and the $t$-distribution with $\nu = n_x + n_y - 2$ degrees of freedom.

$$(\bar{x} - \bar{y}) \pm t_{n_x+n_y-2}\sqrt{S_p^2\left(\frac{1}{n_x}+\frac{1}{n_y}\right)}$$

**Interpreting the Interval:** If the confidence interval **contains zero** (e.g., spanning from a negative number to a positive number), there is no significant difference between the means at that confidence level. If the entire interval is strictly positive or strictly negative, it indicates a significant difference exists.

### Exercise 17B

**1** For each of the following scenarios, state whether a 2-sample $z$-test or a 2-sample $t$-test should be used:

(a) Comparing the means of two normal populations with known variances $\sigma_1^2 = 12$ and $\sigma_2^2 = 15$. **[1 mark]**

(b) Comparing the means of two normal populations with unknown variances, assuming the population variances are equal. **[1 mark]**

**2** A manufacturer produces two types of batteries, Type A and Type B. The lifespans of both types are normally distributed. The population variance of the lifespan of Type A batteries is known to be 1.4 hours$^2$, and for Type B batteries it is 1.8 hours$^2$. A random sample of 40 Type A batteries has a mean lifespan of 45.2 hours. A random sample of 50 Type B batteries has a mean lifespan of 44.1 hours. Test, at the 5% significance level, whether the mean lifespan of Type A batteries is greater than that of Type B batteries. **[6 marks]**

**3** A city planner is comparing the daily commute times for residents of two different cities, City 1 and City 2. A random sample of 80 residents from City 1 has a mean commute time of 32.4 minutes with an unbiased sample standard deviation of 5.2 minutes. A random sample of 100 residents from City 2 has a mean commute time of 34.1 minutes with an unbiased sample standard deviation of 6.1 minutes. Test, at the 1% significance level, whether there is a difference in the mean commute times of the residents of the two cities. **[6 marks]**

**4** An agricultural researcher is testing two different fertilisers, X and Y, to see their effect on the height of a specific plant. The heights, in cm, of a random sample of plants treated with each fertiliser are recorded below:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Fertiliser X** | 12.4 | 13.1 | 11.9 | 14.0 | 12.8 | 13.5 | |
| **Fertiliser Y** | 10.8 | 11.5 | 12.1 | 11.0 | 10.5 | 11.8 | 11.2 |

Assume that the heights of the plants are normally distributed and that the population variances are equal. Test, at the 5% significance level, whether Fertiliser X produces taller plants on average than Fertiliser Y. **[7 marks]**

**5** A psychologist is comparing the reaction times of video gamers and non-gamers. The reaction times, in milliseconds (ms), are assumed to be normally distributed with equal population variances. A random sample of 12 gamers has a mean reaction time of 215 ms and an unbiased sample variance of 420. A random sample of 15 non-gamers has a mean reaction time of 238 ms and an unbiased sample variance of 510. Test, at the 1% significance level, whether gamers have a faster (lower) mean reaction time than non-gamers. **[7 marks]**

**6** A consumer rights group is investigating the sugar content in two competing brands of apple juice, Brand P and Brand Q. The sugar content, in grams per 100ml, is assumed to be normally distributed with equal population variances. A random sample of 10 cartons of Brand P has a mean sugar content of 10.2 g and an unbiased sample standard deviation of 1.1 g. A random sample of 12 cartons of Brand Q has a mean sugar content of 11.5 g and an unbiased sample standard deviation of 1.3 g.

(a) Calculate a 95% confidence interval for the difference in mean sugar content between Brand Q and Brand P ($\mu_Q - \mu_P$). **[5 marks]**

(b) State, with a reason, whether the confidence interval supports the claim that there is a difference in the mean sugar content of the two brands. **[1 mark]**

**7** An automotive magazine is comparing the fuel efficiency, in miles per gallon (mpg), of two car models, Model X and Model Y. The fuel efficiencies are assumed to be normally distributed with equal population variances. A random sample of 8 Model X cars and 10 Model Y cars yielded the following summary statistics:

$$\sum x = 340, \quad \sum x^2 = 14510$$
$$\sum y = 405, \quad \sum y^2 = 16480$$

(a) Calculate the unbiased sample variances for both car models. **[4 marks]**

(b) Test, at the 5% significance level, whether there is a difference in the mean fuel efficiency of the two car models. **[6 marks]**

## 17.3 2-Sample Tests for Proportions

We can also test whether the proportion of 'successes' in two distinct populations is the same. For example, comparing the defect rates from two different factories or the success rates of two different marketing campaigns.

Under the null hypothesis $H_0 : \pi_1 = \pi_2 = \pi$, we assume both samples come from populations with the same underlying proportion $\pi$. We estimate this common proportion by pooling the successes from both samples.

**Key Point**

When testing the difference between two binomial proportions, we use the normal approximation.

- **Pooled proportion estimate:** $p = \frac{x_1+x_2}{n_1+n_2}$
- **Standard error:** $SE = \sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$
- **Test statistic for difference in proportions:**

$$z = \frac{p_1 - p_2}{\text{Standard Error}} \sim N(0,1)$$

Where $p_1 = \frac{x_1}{n_1}$ and $p_2 = \frac{x_2}{n_2}$ are the sample proportions.

**Example 4**

**Question:** Two factories, X and Y, produce the same component. A random sample of 200 components from Factory X contains 12 defective items. A random sample of 300 components from Factory Y contains 27 defective items. Test at the 10% significance level whether the proportion of defective items from Factory Y is greater than from Factory X.

**Solution:**

- **Sample Proportions:**

$$p_X = \frac{12}{200} = 0.06, \quad p_Y = \frac{27}{300} = 0.09$$

- **Hypotheses:** Let $\pi_X$ and $\pi_Y$ be the true population proportions.

$$H_0 : \pi_X = \pi_Y$$

$$H_1 : \pi_Y > \pi_X$$

- **Pooled Proportion:** Under $H_0$, we estimate the common proportion $p$:

$$p = \frac{12 + 27}{200 + 300} = \frac{39}{500} = 0.078$$

- **Standard Error:**

$$\begin{aligned} SE &= \sqrt{p(1-p)\left(\frac{1}{n_X}+\frac{1}{n_Y}\right)} \\ &= \sqrt{0.078(0.922)\left(\frac{1}{200}+\frac{1}{300}\right)} \\ &= \sqrt{0.071916 \times 0.008333\ldots} \\ &\approx 0.02448 \end{aligned}$$

- **Test Statistic:**

$$z = \frac{p_Y - p_X}{SE} = \frac{0.09 - 0.06}{0.02448} = \frac{0.03}{0.02448} \approx 1.225$$

- **Critical Value:** For a 10% one-tailed test, the critical $z$-value is 1.282.
- **Conclusion:** Since $1.225 < 1.282$, we do not reject $H_0$. There is insufficient evidence at the 10% level to conclude that Factory Y produces a higher proportion of defective items than Factory X.

## Exercise 17C

**1** A manufacturer sources electronic components from two different suppliers, Supplier X and Supplier Y. In a random sample of 400 components from Supplier X, 15 are found to be defective. In a random sample of 500 components from Supplier Y, 28 are found to be defective. The manufacturer wishes to test, at the 5% significance level, whether the proportion of defective components from Supplier X is lower than that from Supplier Y.

(a) State suitable null and alternative hypotheses for this test. **[1 mark]**

(b) Calculate the pooled estimate for the common population proportion. **[1 mark]**

(c) Calculate the test statistic. **[3 marks]**

(d) State your conclusion in context. **[2 marks]**

**2** An educational researcher is comparing the first-time pass rates of two driving schools. School A had 112 students pass out of a random sample of 160. School B had 145 students pass out of a random sample of 225.

(a) Carry out a hypothesis test at the 10% significance level to determine if there is a difference in the true first-time pass rates between the two driving schools. **[6 marks]**

**3** A tech company is A/B testing two different designs for their website's landing page to see which generates more user sign-ups. Design A was shown to a random sample of 600 visitors, resulting in 60 sign-ups. Design B was shown to a random sample of 750 visitors, resulting in 120 sign-ups.

(a) Test, at the 1% significance level, whether Design B has a higher sign-up rate than Design A. **[6 marks]**

(b) State the critical value used in your test. **[1 mark]**

**4** ⋆ An ecologist is studying the prevalence of a specific fungus in two separate forests. In Forest 1, a random sample of 200 trees reveals that 40 are infected with the fungus. In Forest 2, a random sample of 250 trees reveals that 80 are infected.

(a) Calculate the $p$-value for a test to determine if there is a difference in the proportion of infected trees between the two forests. **[4 marks]**

(b) Using your $p$-value, state your conclusion at the 5% significance level. **[2 marks]**

**5** A tennis coach is analysing the serving performance of two players. Player A successfully lands 310 first serves out of a random sample of 500 attempts. Player B successfully lands 240 first serves out of a random sample of 450 attempts.

(a) State the hypotheses to test whether Player A has a higher true proportion of successful first serves than Player B. **[1 mark]**

(b) Calculate the test statistic for this hypothesis test. **[3 marks]**

(c) Conclude the test at the 2.5% significance level, interpreting your result in context. **[2 marks]**

**6** A pharmaceutical company is comparing the incidence of nausea as a side effect between two different painkillers. In a clinical trial, 45 out of a random sample of 300 patients taking Drug A experienced nausea. For Drug B, 22 out of a random sample of 250 patients experienced nausea.

(a) Carry out a hypothesis test at the 10% significance level to investigate whether Drug A causes a higher proportion of nausea than Drug B. **[6 marks]**

**7** A political analyst is investigating support for a new environmental policy in two different cities. In City P, 215 out of a random sample of 400 residents support the policy. In City Q, 280 out of a random sample of 550 residents support the policy.

(a) Test at the 5% significance level whether there is a difference in the proportion of residents who support the policy between the two cities. **[6 marks]**

## 17.4 Hypothesis Tests for Variance (1 and 2 Samples)

*Note: This section is beyond the scope of the Edexcel Statistics A-Level course, but it is required for Further Statistics 2 in the Edexcel Further Mathematics specification.*

### 17.4.1 1-Sample Chi-Squared Test for Variance

We can test whether a population variance $\sigma^2$ equals a specific value $\sigma_0^2$ using the Chi-squared distribution.

**Key Point**

To test $H_0 : \sigma^2 = \sigma_0^2$ for a Normal population:

1. **Test Statistic:** $X^2 = \frac{(n-1)S^2}{\sigma_0^2}$
2. **Distribution:** Under $H_0$, $X^2 \sim \chi^2_{n-1}$
3. Compare the test statistic to the critical values from the $\chi^2$ table at the required significance level.

**Example 5**

**Question:** A machine produces metal rods. Historically, the variance of the length of the rods is known to be 0.5 mm$^2$. A random sample of 10 newly produced rods is taken, and the unbiased sample variance is found to be 1.2 mm$^2$. Assuming the lengths of the rods are normally distributed,

test at the 5% significance level whether the variance of the lengths has increased.
**Solution:**

- **Hypotheses:** $H_0 : \sigma^2 = 0.5$ versus $H_1 : \sigma^2 > 0.5$
- **Significance Level:** $\alpha = 0.05$ (One-tailed test)
- **Degrees of Freedom:** $\nu = n - 1 = 10 - 1 = 9$
- **Test Statistic:**
$$X^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{9 \times 1.2}{0.5} = \frac{10.8}{0.5} = 21.6$$
- **Critical Value:** From the $\chi^2$ distribution tables for $\nu = 9$ and an upper-tail probability of 0.05, the critical value is 16.919.
- **Conclusion:** Since the test statistic is greater than the critical value ($21.6 > 16.919$), it falls in the critical region so we reject $H_0$. There is sufficient evidence at the 5% significance level to suggest that the variance of the lengths of the metal rods has increased.

### 17.4.2 2-Sample $F$-test for Equal Variances

Before performing a pooled 2-sample $t$-test, you must assume the two population variances are equal ($\sigma_1^2 = \sigma_2^2$). You can formally test this assumption using the **F-distribution**.

**Key Point**

To test $H_0 : \sigma_1^2 = \sigma_2^2$ versus $H_1 : \sigma_1^2 \neq \sigma_2^2$:

1. **Test Statistic:** Always divide the *larger* sample variance by the *smaller* sample variance.
$$F = \frac{S^2_{\text{large}}}{S^2_{\text{small}}}$$
2. **Degrees of Freedom:** The F-distribution has two sets of degrees of freedom. $\nu_1 = n_{\text{large}} - 1$ (numerator df) and $\nu_2 = n_{\text{small}} - 1$ (denominator df).
3. **Critical Value:** Because we forced $F \geq 1$ by putting the larger variance on top, we only need to look at the **upper tail**. For a two-tailed test at significance level $\alpha$, compare $F$ to the critical value at the $\frac{\alpha}{2}$ level.

If $F >$ Critical Value, reject $H_0$; the variances are not equal.

**Example 6**

**Question:** A researcher is investigating the weights of two different breeds of dog, Breed A and Breed B. A random sample of 11 dogs of Breed A has an unbiased sample variance of 15.2 kg$^2$. A random sample of 9 dogs of Breed B has an unbiased sample variance of 4.1 kg$^2$. Assume the weights of both breeds are normally distributed. Test, at the 10% significance level, whether there is a difference in the population variances of the weights of the two breeds.
**Solution:**

- **Hypotheses:** $H_0 : \sigma_A^2 = \sigma_B^2$ versus $H_1 : \sigma_A^2 \neq \sigma_B^2$
- **Significance Level:** $\alpha = 0.10$. Since it is a two-tailed test, we look at the $\frac{\alpha}{2} = 0.05$ level for the upper tail.
- **Degrees of Freedom:** The test statistic is formed by putting the larger variance in the numerator.
  - Numerator (Breed A, larger variance): $\nu_1 = n_A - 1 = 11 - 1 = 10$
  - Denominator (Breed B, smaller variance): $\nu_2 = n_B - 1 = 9 - 1 = 8$
- **Test Statistic:**
$$F = \frac{S^2_{\text{large}}}{S^2_{\text{small}}} = \frac{15.2}{4.1} \approx 3.707$$
- **Critical Value:** From the $F$-distribution tables for $\nu_1 = 10$ and $\nu_2 = 8$ with an upper-tail probability of 0.05, the critical value is 3.347.
- **Conclusion:** Since $3.707 > 3.347$, the result is significant and we reject $H_0$. There is sufficient evidence at the 10% significance level to conclude that the population variances of the weights of the two breeds of dog are not equal.

## End-of-Chapter Mixed Exercises

1. A machine fills bags of sugar. The weights are known to be normally distributed with standard deviation 4 g. The machine is set to fill bags with a mean of 500 g. A random sample of 10 bags has a mean weight of 497 g. Test at the 5% level of significance whether the mean weight is less than 500 g. **[5 marks]**

2. A coin is suspected of being biased towards heads. It is flipped 40 times and lands on heads 26 times. Test at the 5% significance level whether the coin is biased towards heads. **[5 marks]**

3. ⋆ A pharmaceutical company claims that a new treatment is more effective than the standard treatment, which has a known success rate of 60%. The new treatment is administered to a random sample of 40 patients. Let $X$ be the number of patients successfully cured by the new treatment.

   (a) State the hypotheses for a test of the company's claim. **[2 marks]**

   (b) Find the critical region for the test at the 5% level of significance. **[3 marks]**

   (c) State the actual level of significance of this test. **[1 mark]**

   (d) The new treatment successfully cured 30 of the 40 patients. State your conclusion in context. **[2 marks]**

4. ⋆ An educational researcher is investigating the scores of students on a standardised test in two different regions, Region P and Region Q. A random sample of 50 students from Region P has a mean score of 72.4 and a standard deviation of 8.5. A random sample of 60 students from Region Q has a mean score of 68.9 and a standard deviation of 9.2.

   (a) Carry out a hypothesis test at the 1% significance level to determine if the mean score of students in Region P is significantly greater than the mean score of students in Region Q. **[5 marks]**

   (b) Explain why it is not necessary to assume that the test scores are normally distributed. **[2 marks]**

5. A call centre claims that the mean handling time for customer calls is 6 minutes. The handling time is assumed to be normally distributed. A random sample of 9 calls gives handling times, in minutes, of:

   $$6.4, \quad 5.8, \quad 6.9, \quad 5.5, \quad 6.2, \quad 7.1, \quad 5.9, \quad 6.6, \quad 6.0$$

   Test, at the 5% significance level, whether there is evidence that the mean handling time differs from 6 minutes. **[5 marks]**

6. Two solar panel manufacturers, SolarMax and SunPeak, produce panels whose efficiency (%) is normally distributed. The population variance of efficiency for SolarMax panels is known to be $1.8\%^2$ and for SunPeak panels is $2.1\%^2$. A random sample of 35 SolarMax panels has a mean efficiency of 21.4%. A random sample of 40 SunPeak panels has a mean efficiency of 20.9%. Test, at the 5% significance level, whether SolarMax panels have a higher mean efficiency than SunPeak panels. **[5 marks]**

7. A researcher is comparing the average daily screen time, in minutes, of teenagers and adults. A random sample of 50 teenagers has a mean daily screen time of 245 minutes with an unbiased sample standard deviation of 38 minutes. A random sample of 60 adults has a mean daily screen time of 231 minutes with an unbiased sample standard deviation of 42 minutes. Test, at the 1% significance level, whether there is a difference in the mean daily screen time between teenagers and adults. **[5 marks]**

8. A study compares two revision techniques used by students preparing for an exam: flashcards (Technique 1) and practice tests (Technique 2). The exam scores for both groups are assumed to be normally distributed with equal population variances. A random sample of 11 students using Technique 1 has a mean score of 68.4 and an unbiased sample variance of 42.5. A random sample of 13 students using Technique 2 has a mean score of 74.1 and an unbiased sample variance of 38.9.

   (a) Test, at the 5% significance level, whether Technique 2 produces a higher mean exam score than Technique 1. **[5 marks]**

   (b) State one assumption, other than normality, that is required for this test to be valid. **[1 mark]**

9. A courier company operates two delivery hubs. Delivery times, in minutes, are assumed to be normally distributed with equal population variances. A random sample of 12 deliveries from Hub 1 has a mean delivery time of 34.2 minutes and an unbiased sample variance of 25.6. A random sample of 15 deliveries from Hub 2 has a mean delivery time of 31.8 minutes and an unbiased sample variance of 22.1.

   (a) Calculate a 90% confidence interval for the difference in mean delivery times between Hub 1 and Hub 2 ($\mu_1 - \mu_2$). **[5 marks]**

   (b) Using your confidence interval, state, with a reason, whether there is a significant difference between the mean delivery times of the two hubs. **[2 marks]**

10. A technology reviewer is comparing the crash rates of two smartphone apps. In a random sample of 400 sessions using App A, 18 sessions resulted in a crash. In a random sample of 500 sessions using App B, 35 sessions resulted in a crash. Test, at the 5% significance level, whether App A has a lower crash rate than App B. **[6 marks]**

11. For each of the following scenarios, state which of the tests from this chapter should be used, giving a reason for your choice:

    (a) Comparing the mean fuel consumption of two car engines, where both population variances are known. **[1 mark]**

    (b) Testing whether the mean weight of a single batch of chocolate bars differs from a stated value, where the population variance is unknown and the sample is small. **[1 mark]**

    (c) Comparing the proportion of customers who prefer two different product designs, based on two large independent samples. **[1 mark]**

12. A vineyard claims that the mean volume of wine in its bottles is 750 ml. The volume is assumed to be normally distributed. A random sample of 16 bottles is taken and the volumes, $x$ ml, are summarised as follows:

$$\sum x = 11952, \quad \sum x^2 = 8935220.8$$

    (a) Calculate the unbiased sample variance of the bottle volumes. **[2 marks]**

    (b) Test, at the 1% significance level, whether there is evidence that the bottles are, on average, underfilled. **[5 marks]**

13. A personal trainer is comparing the effectiveness of two weight-loss programmes. The weight lost, in kg, after 8 weeks is assumed to be normally distributed with equal population variances. A random sample of participants on Programme 1 lost the following amounts of weight:

$$4.2, \quad 3.8, \quad 5.1, \quad 4.5, \quad 3.9, \quad 4.8, \quad 4.0, \quad 5.3, \quad 4.4$$

A random sample of participants on Programme 2 lost the following amounts of weight:

$$3.1, \quad 2.8, \quad 3.5, \quad 3.9, \quad 2.6, \quad 3.3, \quad 3.0, \quad 2.9$$

Test, at the 5% significance level, whether there is a difference in the mean weight lost between the two programmes. **[6 marks]**

14. A marketing team compares the open rates of two email newsletters. In a random sample of 600 subscribers sent Newsletter A, 84 opened the email. In a random sample of 650 subscribers sent Newsletter B, 105 opened the email.

    (a) Calculate the $p$-value for a test to determine whether Newsletter B has a higher open rate than Newsletter A. **[4 marks]**

    (b) Using your $p$-value, state your conclusion at the 5% significance level. **[2 marks]**

# Chapter 17 Summary

**Summary of Key Formulas**

1. **1-sample $t$-test statistic:** $t = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$
2. **2-sample $z$-test statistic:** $z = \frac{(\bar{X}-\bar{Y})-(\mu_x-\mu_y)}{\sqrt{\frac{\sigma_x^2}{n_x}+\frac{\sigma_y^2}{n_y}}} \sim N(0,1)$
3. **Pooled variance:** $S_p^2 = \frac{(n_x-1)S_x^2+(n_y-1)S_y^2}{n_x+n_y-2}$
4. **2-sample $t$-test statistic (equal variances):** $t = \frac{(\bar{X}-\bar{Y})-(\mu_x-\mu_y)}{\sqrt{S_p^2\left(\frac{1}{n_x}+\frac{1}{n_y}\right)}} \sim t_{n_x+n_y-2}$
5. **Test statistic for difference in proportions:** $z = \frac{p_1-p_2}{\text{Standard Error}}$
   where Standard Error $= \sqrt{p(1-p)\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$ and $p = \frac{x_1+x_2}{n_1+n_2}$.

# Appendix B

# Use of Calculators

The use of technology is a mandatory requirement of the Edexcel A-Level Statistics specification. You are expected to be highly proficient with your calculator to compute summary statistics, find probabilities from standard distributions, and bypass tedious manual arithmetic.

This appendix provides step-by-step instructions for the three most common calculator models used in the UK:

- **Casio ClassWiz fx-991EX** (The legacy ClassWiz). *Note: This model has been discontinued and replaced by the fx-991CW, but it remains fully permitted in UK public exams.*
- **Casio ClassWiz fx-991CW** (The new ClassWiz)
- **Casio fx-CG50** (Graphical Calculator)

**Key Point**

**Calculator Exam Rules:**
You are allowed to use calculators with advanced statistical distributions in the exam. Graphical calculators (like the CG-50) are permitted, provided they do not have Computer Algebra System (CAS) capability (the CG-50 does not have CAS). Calculators must not have access to the internet or store unapproved files.

## B.1 Generating Random Numbers (Sampling)

In Chapter 3, you learned about simple random sampling. You can use your calculator as a random number generator to select items from a sampling frame.

### Casio fx-991EX

1. Press **ALPHA** then **.** to select `RanInt#`.
2. Enter the lower bound, press **SHIFT** then **)** to type a comma **,**, enter the upper bound, and close the brackets **)**.
3. Example: `RanInt#(1, 50)`. Press **=** to generate a random integer. Press **=** repeatedly for more numbers.

### Casio fx-991CW

1. Press **CATALOG**, select **Probability**, then scroll down to **Random Integer**, and press **EXE**.
2. Enter the lower bound, press **SHIFT** then **)** to type a comma **,**, enter the upper bound, and close the brackets **)**.
3. Example: `RanInt#(1, 50)`. Press **EXE** to generate a random integer. Press **EXE** repeatedly for more numbers.

### Casio fx-CG50

1. From the **Run-Matrix** menu, press **OPTN**, **F6** (▷), **F3** (PROB), **F4** (RAND), **F2** (Int).
2. Type the lower bound, a comma **,** (using the dedicated comma key), and the upper bound.
3. Example: `RanInt#(1, 50)`. Press **EXE** to generate a random integer. Press **EXE** repeatedly for more numbers.

## B.2 Summary Statistics (1-Variable Data)

You will frequently need to find the mean ($\bar{x}$) and sample standard deviation ($s_x$) from raw data or a frequency table.

**Key Point**

**Population vs. Sample Standard Deviation**
Calculators output two different standard deviations. You must choose the correct one!

- $\sigma_x$ (or $\sigma x$): The **population** standard deviation (divided by $n$).
- $s_x$ (or $sx$): The **sample** standard deviation (divided by $n-1$). This is the unbiased estimator used in hypothesis testing (e.g., $t$-tests).

If you need the variance, simply take the standard deviation value and square it!

### Casio fx-991EX

1. **Turn on Frequency (if needed):** Press **SHIFT**, **MENU** (SETUP), press ↓, select **3: Statistics**, select **1: On**.
2. **Enter Data:** Press **MENU**, select **6: Statistics**, select **1: 1-Variable**. Enter your $x$ values (and frequencies if applicable).
3. **View Results:** Press **OPTN**, select **3: 1-Variable Calc**. A list of summary statistics (including $\bar{x}$, $\Sigma x$, $\sigma x$, and $sx$) will appear.

### Casio fx-991CW

1. **Turn on Frequency (if needed):** Press **HOME**, select **Statistics**, select **1-Variable**. Press **TOOLS**, select **Frequency**, select **On**. Press the **Back** key (curved arrow) to return to the data table.
2. **Enter Data:** Enter your $x$ values and frequencies into the table.

3. **View Results:** After entering the last data point, press **EXE**. A menu will appear. Select **1-Var Results**. Scroll down the list to view $\bar{x}$, $\Sigma x$, $\sigma_x$, and $sx$.

### Casio fx-CG50

1. **Enter Data:** Press **MENU**, select **2: Statistics**. Enter your data into **List 1** (and frequencies into **List 2** if applicable).
2. **Setup:** Press **F2** (CALC), then **F6** (SET). Ensure `1Var XList` is set to **List1**. If using a frequency table, set `1Var Freq` to **List2**. If using raw data, ensure it is set to **1**. Press **EXIT**.
3. **View Results:** Press **F1** (1-VAR) to view the summary statistics.

## B.3 Correlation and Linear Regression (2-Variable Data)

### Casio fx-991EX

1. Press **MENU**, select **6: Statistics**, select **2:** $y = a + bx$.
2. Enter your $x$ and $y$ data into the columns.
3. Press **OPTN**, select **4: Regression Calc**. The calculator will display the $y$-intercept ($a$), the gradient ($b$), and Pearson's PMCC ($r$).

### Casio fx-991CW

1. Press **HOME**, select **Statistics**, select **2-Variable**.
2. Enter your $x$ and $y$ data into the columns.
3. After entering the last data point, press **EXE**. A menu will appear. Select **2-Var Results** to view standard summary statistics, or select **Reg Results** to view the regression coefficients ($a$ and $b$) and Pearson's PMCC ($r$).

### Casio fx-CG50

1. Press **MENU**, select **2: Statistics**. Enter your $x$ data into **List 1** and $y$ data into **List 2**.
2. Press **F2** (CALC), then **F3** (REG), then **F1** ($X$), then **F2** ($a + bx$).
3. The screen will display the $y$-intercept ($a$), the gradient ($b$), and Pearson's PMCC ($r$). *(Note: selecting $a + bx$ ensures the variables match the Edexcel formula booklet and ClassWiz notation.)*

**Continue in the full book**

This evaluation copy includes Sections B.1–B.3 only. The complete book contains the rest of the calculator guide, including workflows for probability distributions and statistical inference.

# Practice Paper 02

## Paper 02: Statistical Inference

Pearson Edexcel 9ST0 A Level Statistics Practice

**Time:** 2 hours
**Total marks:** 80

**Instructions**

- Answer all questions.
- A calculator may be used.
- The booklet *Statistical Formulae and Tables* may be used.
- Show sufficient working to make your method clear.
- The marks for each question are shown in brackets.

This is independent practice material.
It is not an official Pearson examination paper and is not endorsed by Pearson.

1. A bakery produces large batches of wholemeal loaves. The target weight for each loaf is 800 grams. To monitor quality, a random sample of 50 loaves is selected from a single day's production. The sample mean weight is found to be 805.2 grams, and the sample standard deviation is 12.4 grams.

   (a) Calculate a 95% confidence interval for the mean weight of all the loaves produced that day. **[4 marks]**

   (b) Interpret the meaning of your confidence interval in the context of the bakery's production. **[2 marks]**

   (c) The head baker claims that the true mean weight of the day's loaves is exactly 800 grams. Use your confidence interval to evaluate this claim at the 5% significance level. **[2 marks]**

   (d) Explain the role of the Central Limit Theorem in the calculation of your confidence interval, and state why it is applicable to this sample. **[2 marks]**

   (e) A trainee suggests that, for tomorrow's sample, they should simply weigh the first 50 loaves that come out of the oven in the morning to save time. Evaluate this proposed data-collection method, identifying a potential source of bias and suggesting an improvement. **[2 marks]**

2. An agricultural researcher is testing two different organic fertilisers, Formula A and Formula B, to determine if there is a difference in the median growth of tomato plants after four weeks. The sample sizes are small, and a preliminary analysis of the data suggests that the distributions of plant growth are heavily skewed.

   Assume the samples are independent and the two population distributions have the same shape, so a rank-sum comparison can be interpreted as a location/median comparison. The recorded growths, measured in centimetres, are as follows:

   | | | | | | | | |
   |---|---|---|---|---|---|---|---|
   | **Formula A:** | 14.2 | 18.1 | 12.5 | 19.0 | 15.3 | 17.6 | 22.4 |
   | **Formula B:** | 9.8 | 11.2 | 13.5 | 10.1 | 16.0 | 8.7 | |

   (a) State why a non-parametric test is more appropriate than a two-sample $t$-test for analysing these data. **[2 marks]**

   (b) Carry out a Wilcoxon rank-sum test (Mann-Whitney $U$ test) at the 5% significance level to determine if there is a significant difference in the median growth of plants treated with Formula A compared to those treated with Formula B. State your hypotheses clearly. **[7 marks]**

   (c) In the context of the statistical enquiry cycle and experimental design, identify one environmental variable that the researcher must control to ensure the validity of this study. **[1 mark]**

3. A city planner is investigating whether there is an association between a commuter's age group and their primary mode of transport. A survey is conducted, and the results are summarised in the contingency table below.

   | | **Car** | **Bus** | **Train** | **Bicycle** |
   |---|---|---|---|---|
   | **Under 30** | 45 | 32 | 28 | 25 |
   | **30 to 50** | 68 | 41 | 50 | 16 |
   | **Over 50** | 52 | 27 | 31 | 9 |

   (a) State the null and alternative hypotheses for a chi-squared test for independence on these data. **[2 marks]**

   (b) Calculate the expected frequency for a commuter who is "Under 30" and uses a "Bicycle". Show your working. **[2 marks]**

(c) State the number of degrees of freedom for this hypothesis test. **[1 mark]**

(d) The planner computes the test statistic to be $\chi^2 = 11.81$. Given that the critical value for this test at the 1% significance level is 16.81, carry out the remainder of the test. State your conclusion in context. **[4 marks]**

(e) The data were collected by interviewing people waiting at a single major train station on a Tuesday morning. Discuss how this sampling methodology might compromise the reliability of the conclusions drawn about all city commuters. **[2 marks]**

4. A sociologist is investigating the relationship between the number of hours individuals spend on social media per week ($x$) and their self-reported anxiety score ($y$, measured on a scale from 0 to 50). Data from a random sample of 18 young adults yields a Pearson product-moment correlation coefficient of $r = 0.534$.

(a) Conduct a hypothesis test at the 5% significance level to determine if there is a positive linear correlation between social media use and anxiety scores in the population. State your hypotheses clearly. **[4 marks]**

(b) The '$p$-value' for this hypothesis test is calculated to be 0.011. Explain exactly what this '$p$-value' represents in the context of the sociologist's test. **[2 marks]**

(c) Calculate the coefficient of determination for these data and interpret its meaning in this context. **[2 marks]**

(d) An article in a local magazine cites this study with the headline: "Social Media Proven to Cause Anxiety in Young Adults!" Critique this headline based on the statistical evidence provided. **[2 marks]**

5. An educational psychologist wishes to compare the effectiveness of three different revision strategies (Method A, Method B, and Method C) on final exam scores. A total of 30 students are randomly and equally assigned to the three methods, meaning there are 10 students in each group. The results are analysed using a one-way Analysis of Variance (ANOVA).

The sum of squares between the groups (SSB) is calculated to be 240, and the sum of squares within the groups (SSW) is 648.

(a) Construct the complete ANOVA table, including the sources of variation, degrees of freedom, mean squares, and the $F$-ratio. **[5 marks]**

(b) Given that the critical value of $F$ at the 5% significance level is 3.35, conduct the hypothesis test and state your conclusion regarding the revision strategies. **[3 marks]**

(c) State two assumptions about the population data that must hold for a one-way ANOVA to be valid. **[2 marks]**

(d) If the test in part (b) leads to the rejection of the null hypothesis, explain briefly how the psychologist could determine which specific pairs of revision strategies have significantly different mean scores. **[1 mark]**

6. A sports scientist is evaluating a new 8-week plyometric training programme designed to improve the vertical jump height of basketball players. The vertical jump heights (in inches) of 8 randomly selected players are measured before and after the programme.

The data are as follows:

| **Player** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **Before** | 24.1 | 26.5 | 22.0 | 28.2 | 25.4 | 23.8 | 27.1 | 21.5 |
| **After** | 25.5 | 27.0 | 21.8 | 29.5 | 26.8 | 24.5 | 28.0 | 23.1 |

Let $d = \text{After} - \text{Before}$.

(a) Calculate the sample mean and sample standard deviation of the differences, $d$. **[3 marks]**

(b) Assuming the differences in vertical jump height are normally distributed, carry out a paired $t$-test at the 5% significance level to investigate whether the training programme significantly increases vertical jump height. **[5 marks]**

(c) Calculate Cohen's $d$ for this paired sample to measure the effect size of the training programme, and interpret its magnitude. **[3 marks]**

(d) Explain what a Type II error would mean in the context of this study. **[2 marks]**

(e) Suggest one way the sports scientist could modify the study design to increase the power of the statistical test. **[1 mark]**

7. A quality control inspector is checking a machine that produces six-sided dice to ensure they are fair. A single die is selected and rolled 150 times. The frequency of each face landing upwards is recorded below.

| **Face** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Observed Frequency** | 18 | 33 | 21 | 19 | 34 | 25 |

(a) Carry out a goodness-of-fit test at the 5% significance level to determine if there is evidence that the die is biased. **[6 marks]**

(b) The '$p$-value' for the test statistic calculated in part (a) is 0.080. Explain how this '$p$-value' supports or contradicts your conclusion in part (a). **[2 marks]**

(c) During the data review stage, the inspector realises the die was rolled by a mechanical arm that always started with the number '6' facing upwards and used the exact same force every time. Explain how this data-collection method might undermine the test of whether the die itself is fair, and suggest how the procedure should be improved. **[2 marks]**

(d) If the inspector were to mistakenly reject the null hypothesis, state the name of this type of error and describe one practical consequence of making this error for the dice manufacturer. **[2 marks]**

**END OF PAPER**

**TOTAL FOR PAPER IS 80 MARKS**

Assume independent flips with constant probability.

3. (a) $H_0 : p = 0.6$, $H_1 : p > 0.6$

   (b) $X \geq 30$

   (c) 0.0352 This is $P_{p=0.6}(X \geq 30)$, the null probability of the rejection region, rather than the nominal 5% level.

   (d) Since 30 is in the critical region, reject $H_0$. There is significant evidence to support the company's claim that the new treatment is more effective.

4. (a) $H_0 : \mu_P = \mu_Q$; $H_1 : \mu_P > \mu_Q$. $z = 2.071$, $p = 0.0192$ Do not reject $H_0$. There is insufficient evidence at the 1% level that Region P has a higher mean test score than Q. Use $z = \dfrac{72.4 - 68.9}{\sqrt{8.5^2/50 + 9.2^2/60}}$ for independent samples. This is a large-sample normal approximation using estimated variances; it requires independent observations and a reasonable normal approximation for each sample mean.

   (b) With independent observations and finite population variances, the CLT can make both sample means approximately normal at these sample sizes (50 and 60), provided severe skewness or influential outliers do not make the approximation poor. A sample size above 30 is a guideline, not a universal guarantee.

5. $H_0 : \mu = 6$; $H_1 : \mu \neq 6$. $t = 1.512$, $p = 0.169$ Do not reject $H_0$. There is insufficient evidence at the 5% level that the mean call-handling time differs from 6 minutes. Use $t = (\bar{x} - 6)/(s/\sqrt{9})$ on 8 degrees of freedom, with both tails and the unrounded raw-data summaries.

6. $H_0 : \mu_M = \mu_P$; $H_1 : \mu_M > \mu_P$. $z = 1.551$, $p = 0.0605$ Do not reject $H_0$. There is insufficient evidence at the 5% level that SolarMax has a higher mean efficiency than SunPeak. Use $z = \dfrac{21.4 - 20.9}{\sqrt{1.8/35 + 2.1/40}}$ for independent samples. The population variances are known and the populations normal.

7. $H_0 : \mu_T = \mu_A$; $H_1 : \mu_T \neq \mu_A$. $z = 1.834$, $p = 0.0667$ Do not reject $H_0$. There is insufficient evidence at the 1% level that teenagers and adults have different mean daily screen times. Use $z = \dfrac{245 - 231}{\sqrt{38^2/50 + 42^2/60}}$ for independent samples. This is a large-sample normal approximation using estimated variances; it requires independent observations and a reasonable normal approximation for each sample mean.

8. (a) $H_0 : \mu_1 = \mu_2$; $H_1 : \mu_2 > \mu_1$. $t = 2.185$, $p = 0.0199$ Reject $H_0$. There is sufficient evidence at the 5% level that practice tests produce higher mean scores than flashcards. Use $s_p^2 = \dfrac{(13-1)38.9 + (11-1)42.5}{22}$ and $t = \dfrac{74.1 - 68.4}{s_p\sqrt{1/13 + 1/11}}$ on 22 degrees of freedom. Assume independent normal samples with a common population variance.

   (b) The samples must be independent and random.

9. (a) $(-0.817, 5.617)$ minutes for $\mu_1 - \mu_2$. Pool on 25 degrees of freedom: $s_p^2 = (281.6 + 309.4)/25 = 23.64$. Use $2.4 \pm t_{25,0.95}s_p\sqrt{1/12 + 1/15}$. Assume independent normal samples with a common population variance.

   (b) Because the confidence interval contains 0, there is no significant difference between the mean delivery times of the two hubs at the 10% significance level.

10. $H_0 : p_A = p_B$; $H_1 : p_A < p_B$. $z = -1.583$, $p = 0.0567$ Do not reject $H_0$. There is insufficient evidence at the 5% level that App A has a lower crash proportion than B. Under equal proportions use $\hat{p} = (18+35)/(400+500)$ and $z = \dfrac{18/400 - 35/500}{\sqrt{\hat{p}(1-\hat{p})(1/400 + 1/500)}}$. Use the one-sided tail specified by the alternative. Assume independent random samples and Bernoulli observations; the pooled expected success and failure counts in each group exceed 5.

11. (a) Use an independent two-sample $z$-test if the population variances are known and

the mean difference has a normal distribution (exactly for normal populations, or approximately with suitable large independent samples).

(b) Use a one-sample $t$-test provided the small independent sample comes from a normal population; the unknown population variance is estimated by the sample variance.

(c) 2-sample $z$-test for proportions, because we are comparing proportions from two large independent samples.

12. (a) 471.787 $\bar{x} = 11952/16 = 747$ and $s^2 = (8935220.8 - 11952^2/16)/15$.

(b) $H_0 : \mu = 750$; $H_1 : \mu < 750$. $t = -0.552$, $p = 0.2944$ Do not reject $H_0$. There is insufficient evidence at the 1% level that the mean wine volume is below 750 ml. Use $t = (747 - 750)/(s/\sqrt{16})$ on 15 degrees of freedom, with the lower tail and unrounded $s$.

13. $H_0 : \mu_1 = \mu_2$; $H_1 : \mu_1 \neq \mu_2$. $t = 5.583$, $p = 0.0001$ Reject $H_0$. There is sufficient evidence at the 5% level that mean weight loss differs between the programmes. Use $s_p^2 = \dfrac{(9-1)s_1^2 + (8-1)s_2^2}{15}$ and $t = \dfrac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/9 + 1/8}}$ on 15 degrees of freedom. Assume independent normal samples with a common population variance.

14. (a) $p = 0.1441$ Under equal proportions use $\hat{p} = (105 + 84)/(650 + 600)$ and $z = \dfrac{105/650 - 84/600}{\sqrt{\hat{p}(1-\hat{p})(1/650 + 1/600)}}$. Use the one-sided tail specified by the alternative. Assume independent random samples and Bernoulli observations; the pooled expected success and failure counts in each group exceed 5.

(b) Since the $p$-value is greater than 0.05, do not reject $H_0$. There is insufficient evidence to conclude that Newsletter B has a higher open rate than Newsletter A.

# Paired Tests

## Exercise 18A

1. For the exact small-sample paired $t$ procedure, the pair differences should be independent observations from a normal population. The original measurements within each pair need not be independent.

2. (a) Let $d$ be Before minus After. Test $H_0 : \mu_d = 0$ against $H_1 : \mu_d > 0$, using $t = \bar{d}/(s_d/\sqrt{n})$ on 7 degrees of freedom. $t = 2.326$, $p = 0.0265$, reject $H_0$. There is sufficient evidence at the 5% level that the drink reduces mean reaction time. Assume independent pairs and normally distributed differences.

(b) $H_0 : \mu_d = 0$, $H_1 : \mu_d > 0$ (where $\mu_d$ is the population mean difference, Before − After).

3. (a) Mean = 1.4, standard deviation = 1.517. The differences are $2, 2, -1, 1, 3$; $\sum d = 7$, $\sum d^2 = 19$, so $s_d = \sqrt{(19 - 7^2/5)/4}$.

(b) $H_0 : \mu_d = 0$; $H_1 : \mu_d > 0$. $t = 2.064$, $p = 0.054$, do not reject $H_0$. Here $d$ is After minus Before. Assume the differences are independent between clients and normally distributed. There is insufficient evidence at the 5% level that the programme increases mean push-ups. The test has $5 - 1 = 4$ degrees of freedom.

(c) Pairing controls for between-client differences. When before and after scores are positively correlated, it can reduce the standard error and improve power; a power advantage is not guaranteed for every population.

4. (a) Let $d$ be With Machine minus Without Machine. Test $H_0 : \mu_d = 0$ against $H_1 : \mu_d > 0$. Using $t = \bar{d}/(s_d/\sqrt{6})$ on 5 degrees of freedom gives $t = 2.825$, $p = 0.0185$. Reject $H_0$: there is sufficient evidence at the 5% level that the machine increases mean deep-sleep duration. Assume independent pairs and normally distributed differences.

(b) 2.015